CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (EE) Examination

May 2014

EE404 Energy Management and Conservation

Date: 22.05.2014, Thursday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain the energy scenario of India. [07]

Q - 2 (a) Explain Need for Electrical Load Management and Step By Step Approach for Maximum Demand Control. [07]

(b) Explain the national energy strategies and national energy plan. [04]

(c) Draw flow chart of energy planning. [03]

OR

Q - 2 (a) What is Energy? Explain energy science & energy Technology. [07]

(b) Draw the single line diagram of electrical distribution system. [03]

(c) What is energy management cell? Explain its objective. [04]

Q - 3 (a) Explain performance evaluation of typical furnace. [07]

(b) Explain the seven principles of energy management. [07]

OR

Q - 3 (a) Explain energy conservation in any production industry. [07]

(b) Draw the structure of energy intensive organization. [02]

(c) Explain the benefits of energy management. [05]

SECTION – II

Q - 4 (a) Write the Potential Energy Conservation Opportunities in lighting System. [07]

Q - 5 (a) Explain need of cogeneration and principle of cogeneration. [07]

(b) Explain energy Audit of electrical system and Illumination system [07]

OR

Q - 5 (a) Explain environment pollution due to energy use. [07]

(b) Explain energy requirement and energy costing. [03]

(c) From energy conservation Act,2001, list four important duties of an energy manager. [04]

Q - 6 (a) Write the list of electrical system optimization and explain any two in details. [07]

(b) Write 8 points for Potential Energy Conservation Opportunities in motor and [04]

(c) transformer. [03]

Explain energy cost of food.

OR

Q - 6 (a) Explain detailed energy Audit. [07]

(b) Explain overhead and underground pumped hydroelectric energy storage system. [07]

****